

CROSS-LAYER SEEDED FAULT INJECTION ARCHITECTURE

Stress-Testing Every Architectural Layer to Verify Deterministic MCU Safety Containment

LAYER 1: SENSORY & TRANSDUCTION

- Dropped MIPI frames (> 100 ms)
- Blinding glare / sudden lux drop
- PTP clock skew (> 5 ms)

LAYER 2: COMPUTE, BUS & MEMORY

- Linux kernel panic / thread crash
- Synthetic DMA bus saturation
- malloc allocation stalls & page faults

LAYER 3: MODEL & ALGORITHMIC

- Hallucinated 3D bounding boxes
- Out-of-workspace trajectories
- Discontinuous acceleration jumps

LAYER 4: ELECTRICAL & PHYSICAL

- Motor over-temp ($> 105^{\circ}\text{C}$)
- Battery voltage sag (< 10.5 V)
- Encoder line break / phase loss

APPLICATION PROCESSOR (Linux MPU · Sys 2/1.5)

- Runs Vision Transformers, Deliberation & Action Chunking
- Untrusted, stochastic, prone to crashes and latency tails
- Emits candidate trajectory buffer p_t over shared IPC link

Candidate p_t

REAL-TIME SAFETY ENFORCER (MCU · Sys 1)

1. Control Barrier Function Invariant ($h(x) \geq 0$):

Projects candidate p_t onto forward invariant safe set C

2. Dynamic Stopping Clearance Check ($d_gap > d_stop$):

Vetoes proposals exceeding braking envelope

3. Zero-Software Hardware Watchdog Monitor:

Trips Category 1 dynamic stop if MPU heartbeat > 50 ms

Permitted u_t / E-Stop

PHYSICAL PLANT & ACTUATION ($W_t \rightarrow W_t+1$)

Closed-loop motor current loops · Mechanical brakes · Verified safe stop

✓ PASS CRITERION

100% Containment Rate:

- Zero collisions ($d_gap > 0$)
- Bounded stopping time
- Controlled halt < 50 ms

Zero Uncontained Escapes:

- Any single uncontained fault
- trips a REFUSE verdict.

Hardware Watchdog:

- Dedicated NMI timer operates
- independently of Linux OS.